



SEE-R

Mon-38

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Introduction



Problem Statement

Kimberly has challenges navigating unfamiliar environments due to her limited vision and hearing abilities.

Current Challenges

Relies on assistive technology through her phone, which is inadequate.

Kimberly's Needs

She requires a hands-free assistive device that improves her awareness through auditory, visual and tactile feedback.

Must Haves

Comfortable and Lightweight Design
Hands-free Mobility
Long Battery Life

Goal

Developing an innovative solution that allows her to move freely without restricting the use of her hands, making her daily tasks easier and improving her quality of life.



THREE BUTTONS

Power On/Off

Speaker/Vibration

Temperature/Distance

Key Features

- **Dual Sensor System** Temperature/Distance
 - Infrared Sensor – Detects Heat
 - Ultrasonic Sensor – Measure Distance
- **Dual Output Modes** Speaker/Vibration
 - Vibration Mode – Alerts user when near an object or when near a heat source
 - Speaker Mode – Announces distance to an object or the detected temperature.
- **Lightweight & Comfortable Design**
- **All-Day Battery Life**
- **Hands-Free Operation** – Doesn't Restrict Hand Movement

Design Benefits



Ergonomic

SEE-R is designed to be as compact as we can possibly make it. It's design has been refined throughout many iterations to attempt to perfect and minimize its size and weight.



Versatile

SEE-R has a range of features blended together to aide in any scenario, without the need of any additional hardware or software. It's modularity is what makes it special.



Ease of Use

SEE-R is made to be accessible. Its primary function is to aid people with as little effort as possible, which is why SEE-R was made to be easy and quick to learn.



Alternatives



Traditional IR Heat Gun

Whilst this may seem like a great option at first, it actually fails our main constraint with our project being the free use of hands^[1]. A Heat Gun would be inaccessible to those who cannot read it, and also be cumbersome if carried around for prolonged use.



Apple "Measure" App

While this option allows for distance measuring on an already accessible device (a phone), the app itself is often unintuitive and time-consuming to get a direct measurement out of – which SEE-R can provide in less than a second^[3].



Laser Distance Meter

This option is great for a more precise measurement, but is not at all convenient. It's very specialized toward the task it does, which makes it unusable for Kimberly's scenario and it lacks rapid output and audio, unlike SEE-R^[2].

Fabrication Details



Materials Used

Material
9V Battery
9V Battery Adapter
Velcro Straps
PLA Filament/Case
Super Glue

Printing Time

Part	Time
Final Assembly	5 hours 12 mins
Prototype Pieces	4 hours 21 mins
Total	9 hours 33 mins

Cost Breakdown

Material	Cost
9V Battery	\$4.45 ^[4]
9V Battery Adapter	\$17.18 ^[5]
Velcro Straps	\$12.98 ^[6]
Speaker	\$6.40 ^[7]
Ultrasonic Sensor	\$11.55 ^[8]
IR Sensor	\$22.99 ^[9]
Vibration Motor	\$12.79 ^[10]
Total	\$88.34

Fabrication Details

Justification of Building Materials and Construction

For the casing, PLA was used since it was one of the easiest materials to manufacture the required shapes such as the curvature of the base. This is because PLA can be molded using a 3D printer. Other plastics considered included were polypropylene, since it is food safe and heat resistant.

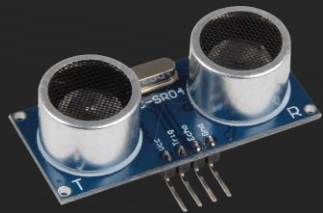
Super glue was used for some of the assembly since a majority of the components in the casing are meant to be permanently bonded to one another. The rest of the case assembly relies on 3D printed fasteners and friction fitted parts for ease of use.

Arduino Circuitry

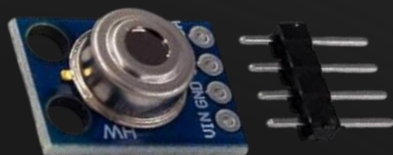
Components	Connections
Ultrasonic Sensor	VCC: 5V GND: GND TRIG: Pin 4 ECHO: Pin 5
IR Sensor	SDL: SDL SCL: SCL VIN: 5V GND: GND
Vibration Motor	VCC: 5V GND: GND IN: Pin 6
Speaker	SIG: Pin 3 GND: GND VCC: 5V
Push Buttons	Input Button: Pin 8 Output Button: Pin 9 Power Button: Pin 10



Arduino-Uno



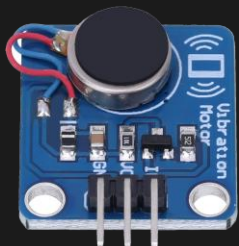
Ultrasonic Sensor



IR Sensor
(MLX90614)

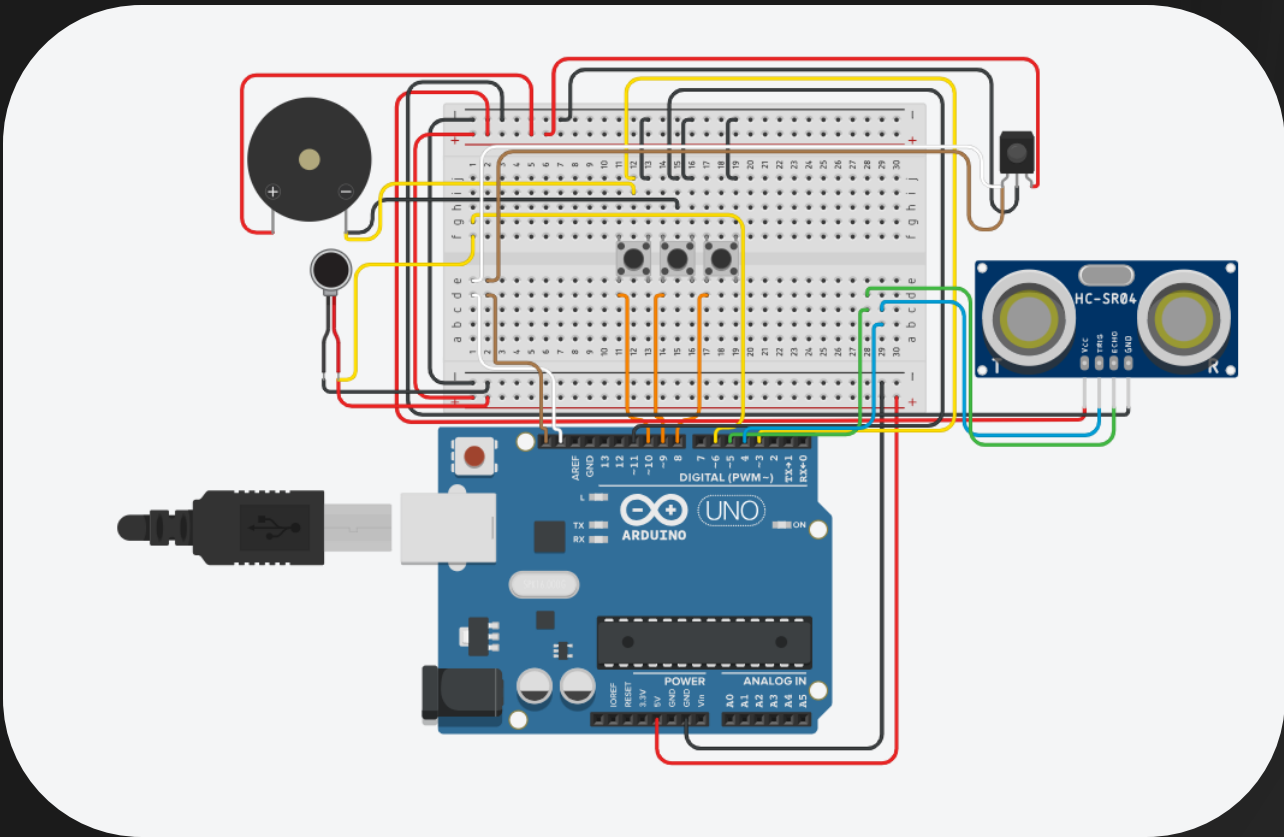


Grove Speaker
(LM386)

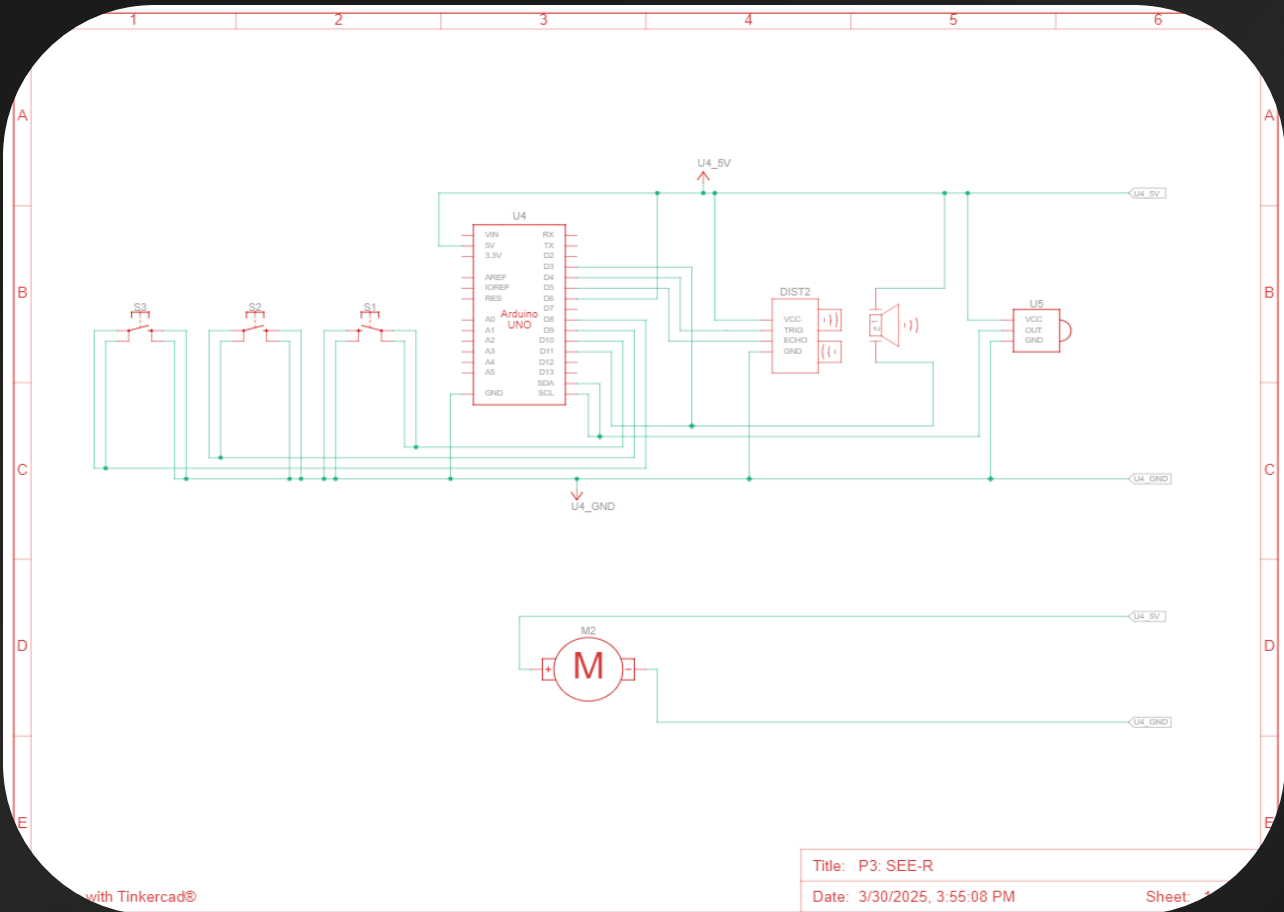


Vibration Motor

Tinker Cad Model



Schematic View



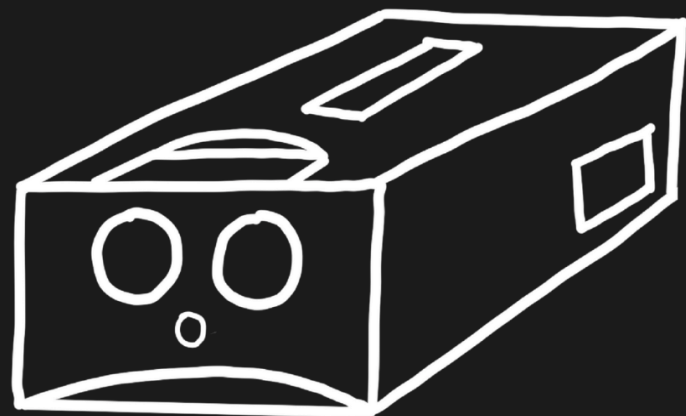
Iterations



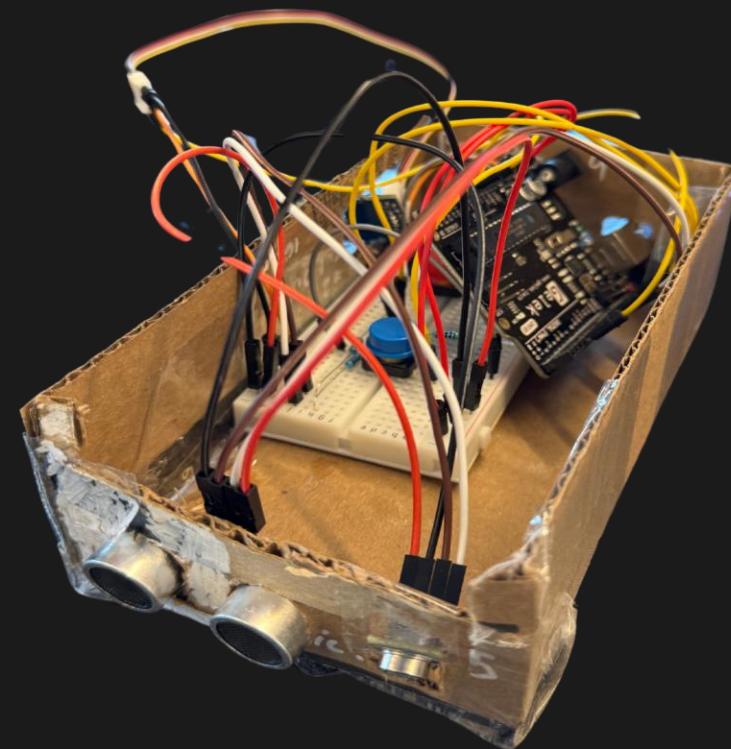
Iterations



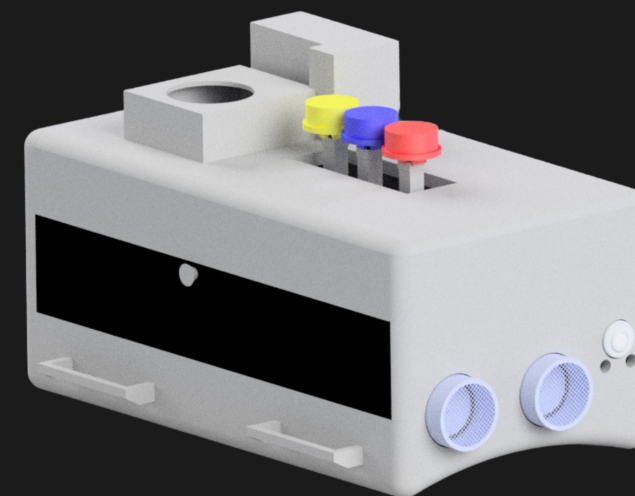
Initial Sketch
and CAD



First
Prototype



Refined Design



Final Device



Testing Results



Battery Life

8-10 hours

SEE-R is powered by a 9V battery. The battery can be replaced easily by removing the back case cover.

Weight

Casing – 103g
Arduino Components – 149g
Total Weight – 252g

13 x 7 x 5.8 cm

The casing contributes to around over 41% of the design's weight and the Arduino components contribute to 59% of the weight.

Comfort Results

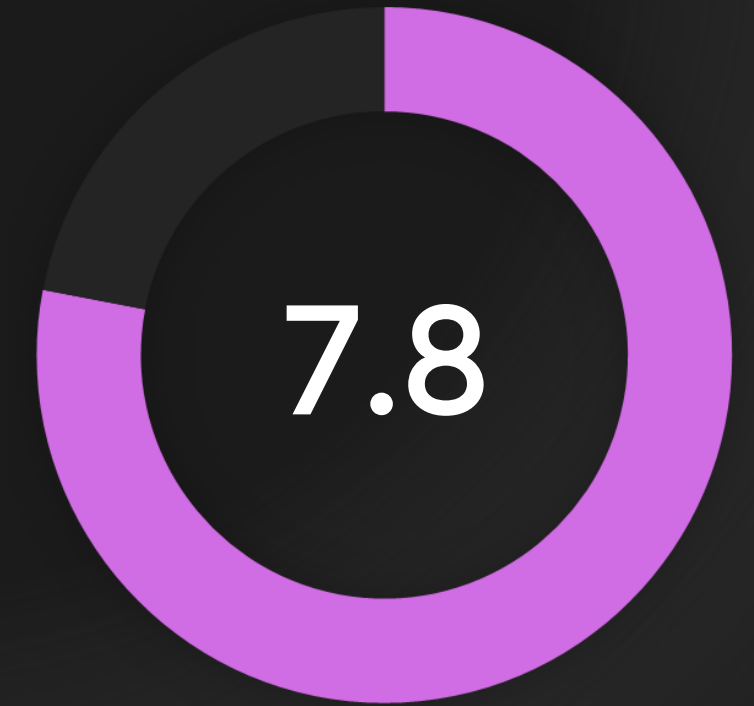
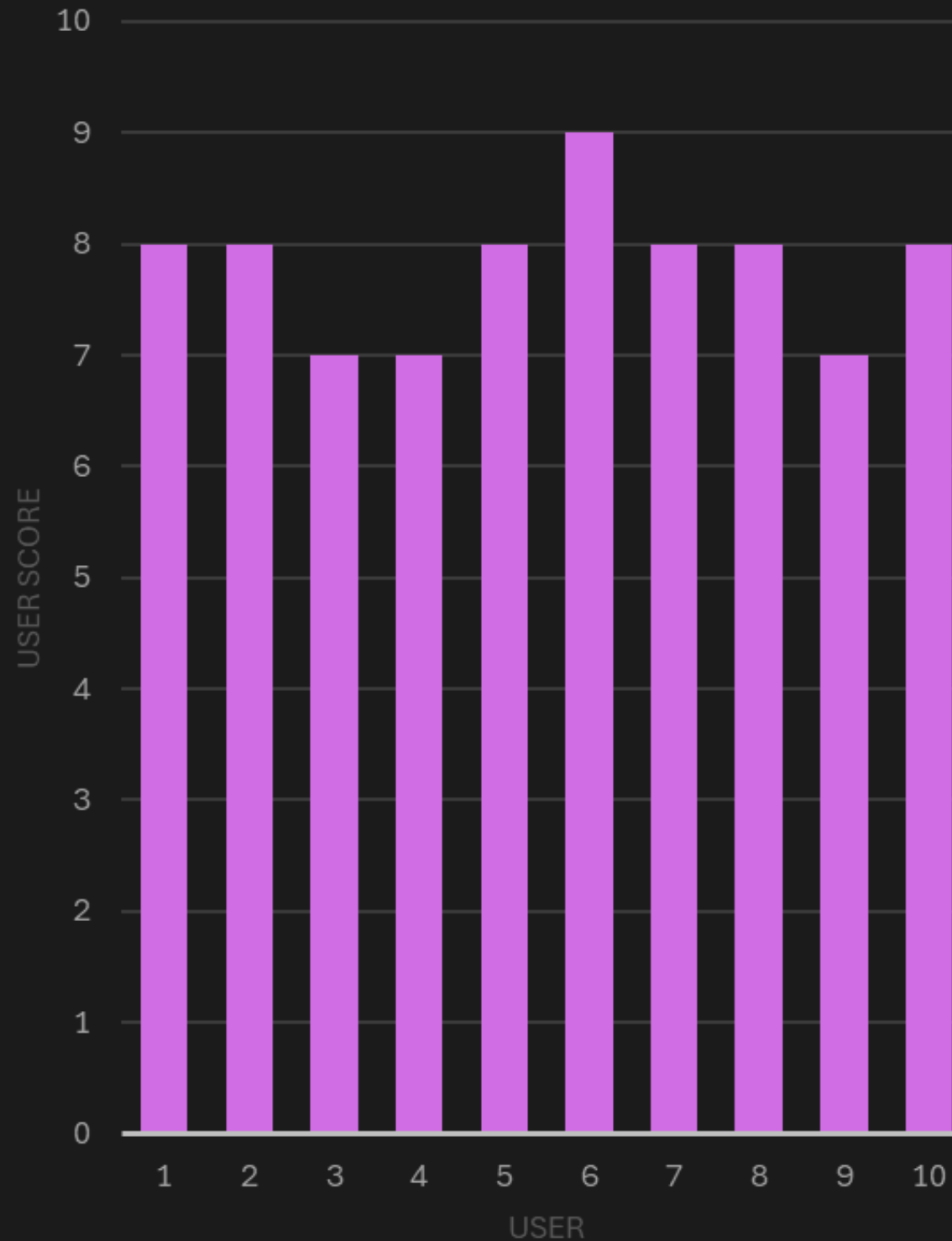


Results

Our design surpassed our expectations in comfort and performed greatly in user scores, achieving an average of 7.8/10 comfort score with most users voting it an 8. This data is vital to our design due to the nature of the product including prolonged usage. Hence, the priority of comfort.

Improvements

Our design is far from its final iteration and can be improved upon. Size could be reduced by using more compact wiring and a specialized processing unit. Furthermore, its comfort can be improved by adding pads or refining the material for a smoother finish.



Functionality Results

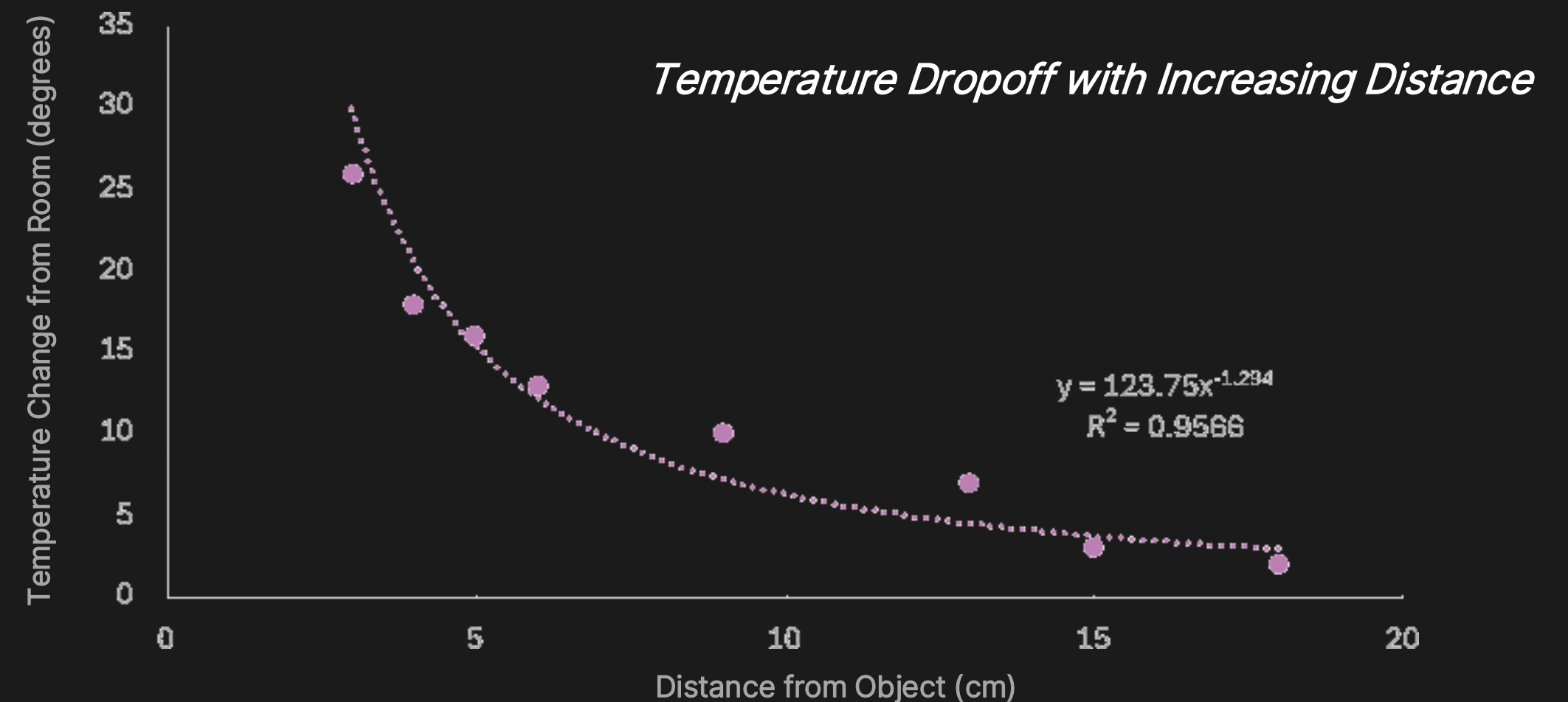
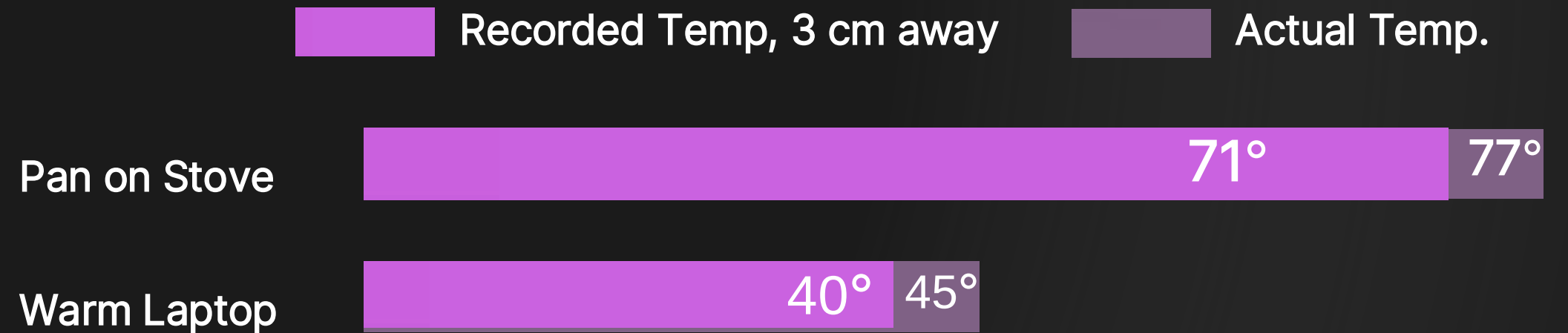


Results

The temperature sensor is able to successfully warn the user of a dangerously hot surface within a safe distance of the object itself. Furthermore, the recorded temperature values are accurate to within ~10 percent of the real value. The distance sensor works flawlessly between a range of 2-100 centimeters.

Improvements

There is a noticeable dropoff of temperature accuracy as the device is moved further away from the object – this can be accounted for with a predictive model that uses both temperature and distance readings. The speaker used to output temperature could be improved with a bigger budget to produce a more audible sound.



References

[1] “Laselion LT50 laser distance meter – 50m (165ft) range, high accuracy, multiple measurement modes, built-in level bubble, and mute function : Amazon.ca: Tools & Home Improvement,” LASELION LT50 Laser Distance Meter – 50m (165ft) Range, High Accuracy, Multiple Measurement Modes, Built-in Level Bubble, and Mute Function : Amazon.ca: Tools & Home Improvement, https://www.amazon.ca/LASELION-LT50-Laser-Distance-Built/dp/B0DHGFXXRY/ref=sr_1_4_sspa?crid=2U7QESNQMUX61&dib=eyJ2IjoiMSJ9.oVgPWTeZwRpNFmAPra6m3fwuratHc6IjEyWiWbCuZ7Ho9bsRNNtxP1f_KQrsgv6tUN2-YdYUkNkKUpRNad4tjUnj1dXCzo9sFYNTv0kKLg7OnBSfihXIKJzVnzGclLK_K6p-ZdzOMzZyW8TFPJb_WBSlilzCVIJO3xD1ND8KIvI0YVvqUjcdI1Q9CMbJth6uJ8giOH6hkbbjDs2FS6B03Epw7GL9L2WWWJE2RWXaY0snyROX8-Hz7kMcCDRDCF7z_6JlujhLGh9gQw6EK8MKKosxhorhpryHuJT-xK73KgE.uSXq7krBi7Py4M5oc03PAXjX8IRfZCtXW_ohs4BXxx8&dib_tag=se&keywords=distance%2Breader&qid=1743364726&spre fix=distance%2Breader%2Caps%2C83&sr=8-4-spons&sp_csd=d2lkZ2V0TmFtZT1zcF9hdGY&th=1+%E2%80%8B (accessed Mar. 30, 2025).

[2] “Etekcity Lasergrip 1080 laser thermometer digital infrared thermometer temperature gun for kitchen cooking BBQ grill and Bath Water, -58°F~1130°F (-50°C~610°C), yellow and black,” Amazon.ca: Industrial & Scientific, https://www.amazon.ca/Etekcity-Lasergrip-1080-Thermometer-Temperature/dp/B00DMI632G?ref_=Oct_d_otopr_d_11853324011_1&pd_rd_w=qqXul&content-id=amzn1.sym.c8e7e75e-d76e-4f67-bacf-4e7f5fa2cf7b&pf_rd_p=c8e7e75e-d76e-4f67-bacf-4e7f5fa2cf7b&pf_rd_r=HKY059KQ3V27MZ5ZHYPX&pd_rd_wg=Qksic&pd_rd_r=b32b9782-2b7a-402f80aa-8bc717613085&pd_rd_i=B00DMI632G&th=1+%E2%80%8B (accessed Mar. 30, 2025).

[3] Apple, "Use the Measure app on your iPhone, iPad, or iPod touch." <https://support.apple.com/en-ca/102468> (accessed Mar. 30, 2025).

[4] “Energizer MAX 9V Battery,” *Walmart.ca*, 2025. <https://www.walmart.ca/en/ip/energizer-max-9v-batteries-1-pack-9-volt-alkaline-batteries/205004?classType=REGULAR&athbdg=L1102&from=/search> (accessed Mar. 30, 2025).

[5] “4pcs Robojax 9v Battery Power Cable for Arduino : Amazon.ca: Electronics,” *Amazon.ca*, 2025. https://www.amazon.ca/dp/B074JNDPF7?ref=ppx_yo2ov_dt_b_fed_asin_title (accessed Mar. 30, 2025).

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[8] “Soldered GY-906 MLX90614 BAA Non-Touch Infrared Temperature Sensor,” *Amazon.ca*. <https://www.amazon.ca/gp/product/B0CNM72R4L?smid=A1XEC9TMFJSNSW&psc=1> (accessed Mar. 30, 2025).

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[10] “FULARR 30Pcs Premium Cable Ties Set.” <https://www.amazon.ca/FULARR-Reusable-Multipurpose-Organizer-Management/dp/B08F7X3TZM>
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